# ThetaSwap: Adverse Competition Oracle

# The Risk Nobody Hedges

In a symmetric equilibrium, each of $N$ LPs earns $1/N$ of fees [Ma and Crapis, 2024]. In practice, sophisticated actors concentrate fee revenue away from passive participants. This is **adverse competition** — a distinct, unhedged risk.

- **Impermanent loss (IL)** — price divergence between pool assets. Variance perpetuals perfectly hedge IL [Fukasawa et al., 2023]; Panoptic deploys this on-chain.
- **Loss-versus-rebalancing (LVR)** — adverse selection cost from informed traders. MEV-aware products address LVR.
- **Fee concentration risk** — the *competition structure* among LPs for fee revenue, captured by the FLAIR metric [Milionis et al., 2024]. **Nothing hedges this today.**

# Research Question

**Is there empirical demand for a hedging instrument against adverse competition? If so, how much are LPs willing to pay?**

## Data

ETH/USDC 30bps pool (Uniswap V3 mainnet)
41 days · 600 positions · 3,365 position-day observations · 597 exits

**Treatment variable — concentration deviation:**

$$\boxed{\Delta^+ = \max\left(0,\ A_T - A_T^{1/N}\right)}$$

- $A_T = \left(\sum_k \theta_k \cdot x_k^2\right)^{1/2}$ fee concentration index ($x_k$ = fee share, $\theta_k = 1/\text{blocklife}$)
- $A_T^{1/N} = \sqrt{\Theta/N^2}$ Ma–Crapis competitive null (equal shares $x_k = 1/N$)
- $\Delta^+ > 0$: fees more concentrated than the symmetric equilibrium

# Model: The Shelter Component

$$P(\text{exit}_{i,t}) = \sigma\big(\beta_0 + \beta_1 \boxed{\Delta_{t-\ell}} + \beta_2\, \Delta^2_{t-\ell} + \beta_3\, \text{IL}_t + \beta_4 \log(\text{age}_{i,t})\big)$$

**We expect** $\beta_1 < 0$ (shelter effect):

- Mild fee concentration correlates with *high trading volume* — the pool is active, all LPs benefit from elevated fee revenue
- At low $\Delta$, more concentration means more volume, which *lowers* exit probability
- This is **healthy competition**: sophisticated LPs attract flow that benefits everyone [Capponi et al., 2024]
- A linear-only model would conclude concentration is *protective* — missing the adverse regime entirely

# Model: The Adverse Competition Component

$$P(\text{exit}_{i,t}) = \sigma\big(\beta_0 + \beta_1\,\Delta_{t-\ell} + \beta_2\,\boxed{\Delta^2_{t-\ell}} + \beta_3\,\text{IL}_t + \beta_4\log(\text{age}_{i,t})\big)$$

**We expect** $\beta_2 > 0$ (adverse competition):

- Beyond a threshold, concentration no longer signals volume — it signals *fee extraction* by sophisticated actors
- The quadratic term $\Delta^2$ captures this **nonlinear regime shift**: the marginal effect of concentration *flips sign* at the turning point
- Together: $\beta_1 < 0 + \beta_2 > 0 =$ **inverted-U** with turning point $\Delta^* = -\beta_1/(2\beta_2)$

# Empirical Results

| Lag | $\hat{\beta}_1$ (linear) | $p$ | $\hat{\beta}_2$ (quadratic) | $p$ | Turning point |
|---|---|---|---|---|---|
| **1** | $\mathbf{-23.18}$ | **0.012** | $\mathbf{+129.20}$ | **0.030** | **0.090** |
| *2* | $-43.42$ | *0.016* | $+226.92$ | *0.065* | *0.096* |
| **3** | $\mathbf{-32.44}$ | **0.001** | $\mathbf{+205.34}$ | **0.004** | **0.079** |

For detailed data curation, specification tests, and robustness checks, see `research/model/main.pdf`.

# What the Data Says

*Passive LPs respond to concentration shocks within one to three days. When fee dispersion exceeds ∼9% above the competitive equilibrium ($\Delta^* = -\beta_1/2\beta_2 = 23.18/(2 \times 129.20) = 0.090$), the shelter effect reverses: each additional percentage point of concentration raises exit probability by approximately 2.3 percentage points ($(\beta_1 + 2\beta_2 \cdot 0.15)\,\bar{P}(1-\bar{P}) = 15.58 \times 0.146 = 2.27$). At a fee dispersion level of 15% above equilibrium, the implied willingness to pay for protection is* **$110 per position per event** *($2.3pp \times 48h \times \$100/h = \$110$).*

This establishes both the **existence of demand** (the inverted-U with a statistically significant adverse regime) and a **concrete price** (the marginal effect translated to an insurance premium).

# Answers

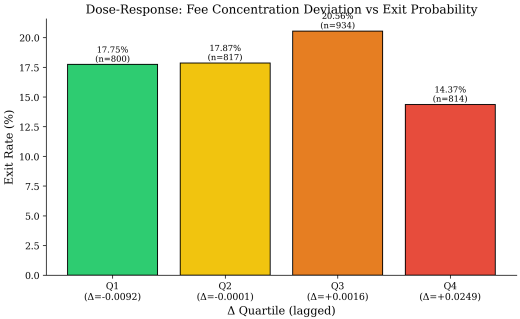
| Question | Finding |
|---|---|
| **Is there demand for hedging adverse competition?** | **Yes.** LPs react to concentration shocks within days. Below a threshold, concentration is protective; above it, adverse competition drives exits with high statistical significance. |
| **What is the price?** | When fee dispersion exceeds 15% above equilibrium, the implied premium is **$110 per position** — the willingness to pay derived from the exit hazard marginal effect. |
| **Does hedging improve outcomes?** | Historical replay with a power-squared payoff and $200K seed capital: **18.8%** of positions are better off hedged, with a mean hedge value of **+$23.21**. |

# Tail Risk: The Inverted-U

The dose-response plot reveals the nonlinear structure:

- Exit rates are **stable at baseline** when concentration is at or below the competitive null (Q1–Q2)
- They **spike** at moderate concentration (Q3) — the adverse regime activates
- At extreme concentration (Q4), exits drop — these rare events are driven by volume spikes that benefit all LPs

This is the signature of a **classic insurable tail risk**: rare, severe, and distinct from price-driven losses.

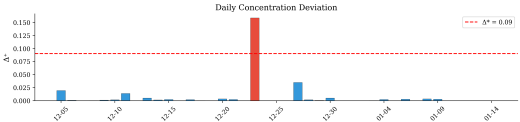


# Tail Risk Confirmation

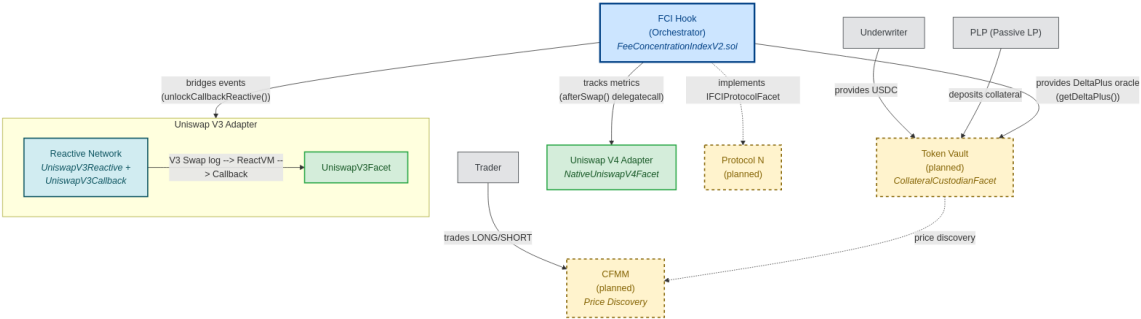
Daily concentration deviation over 41 days:

- 63% of days have $\Delta^+ > 0$ (some concentration)
- **Only 1 day in 41** exceeds the turning point $\Delta^*$
- On that day, $\Delta^+$ jumped from near-zero to **0.158**

**Rare but severe** — the adverse regime is a tail event. A small, predictable premium protects against low-probability, high-impact concentration spikes.

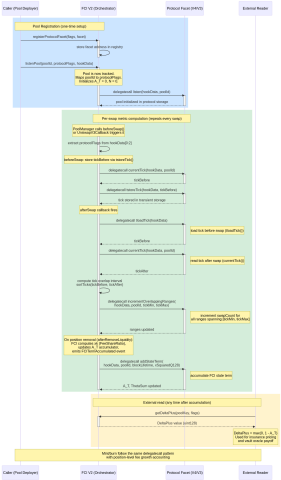


# Architecture




FCI Hook at center — protocol adapters (V3, V4) feed swap/mint/burn events, Reactive Network bridges metrics cross-chain, Vault + CFMM settle downstream.

# Cross-Protocol Modularity




`listenPool()` → swap/mint/burn events → metric update → $\Delta^+$ derivation.

# Live Demo

## Run the integration test

```
forge test --match-path
"test/fee-concentration-index-v2/protocols/uniswapV4/
NativeV4FeeConcentrationIndex.integration.t.sol" -vv
```

**What to observe:**

1. Swap updates $A_T$ — fee concentration index tracks each trade
2. Mint increments $N$ — new positions enter the pool
3. Burn triggers $\Delta^+$ derivation — exit computes concentration deviation
4. All metrics computed on-chain in a single V4 hook callback

# What's Next

**Built**

- FCI oracle live on Uniswap V4
- Reactive adapter bridging V3 events cross-chain
- Backtest validation: 18.8% of positions better off hedged, +$23 mean hedge value

**Next**

- Linearized power-squared CFMM (trading function complete in spec)
- Vault settlement mechanism

*The mathematical specification is complete, the oracle tracks metrics across protocols, and the next milestone delivers the full insurance product.*

# References I


A. Capponi, R. Jia, and R. Zhu. The economics of just-in-time liquidity. *Working paper*, 2024. JIT LP fee dilution and realized fee share asymmetry.

M. Fukasawa, B. Maire, and M. Wunsch. Weighted variance swaps and impermanent loss. *arXiv preprint*, 2023. Variance perpetuals perfectly hedge IL; `https://arxiv.org/abs/2212.08323`.

Y. Ma and D. Crapis. Decentralized exchanges: The uniswap amm. *Working paper*, 2024. Symmetric equilibrium $x_k = 1/N$ as competitive null hypothesis.

J. Milionis, X. Wan, and A. Adams. Flair: A metric for liquidity provider competitiveness in automated market makers. *Working paper*, 2024. Fee-adjusted LP returns metric capturing adverse competition among LPs.